The ECBench ECDLP Performance Frontier

End-to-end boundary, implementation census, bottlenecks, and the next isolated campaign

ECBench frontier audit

1 November 2026

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1 Executive answer

The repository does **not** have one global ECDLP performance frontier. It has several frontiers indexed by curve family, target law, advice model, accounting unit, and evidence class. Once those domains are kept separate, the boundary is clear:

1. **Generic prime, random-binary, and medium/scaling Koblitz elliptic curves remain rho-led.** No admitted cold, one-target index-calculus result in those panels is below its strongest applicable matched generic reference. The closest current repository markers are 2.28× rho for inherited F4 on one tiny random-binary cell, 2.59× for the prime residual collector at 36 bits, and 3.43× on the admitted Koblitz ladder. These are different studies and units, so they are boundary markers, not entries in one sortable table. A separate n=29 one-column generator-orbit panel does cross below rho-frob, but at one column it is a fixed-setup Frobenius-folded BSGS; the gap grows with r, so it is not a uniform-target or asymptotic IC crossing. The late-main prime-orbit relation collector is also rho-led in its new count-first cross-checks: its tuned affine arm costs 5.42× negation rho on the generic shape, 4.20× cubic rho on j=0, and 5.79× quartic rho on j=1728. A late N83 root-index loop also completed two alternating K1107 plain-four-byte/FP2 full-process pairs. FP2 was ahead in both observed whole-process ratios (plain/FP2 1.017 and 1.294), but the shared host load varied sharply and the K5500 Tier-1 comparison remains unmeasured; this is an unranked implementation diagnostic, not a new Koblitz frontier row.
2. **One restricted curve family is IC-led.** The plane-quartic attack on the special GHS-weak family $E_0/\mathbb{F}_{q^3}$, with $\#E_0(\mathbb{F}_{q^3}) = 4\ell$, is below matched rho in its own cold, one-target comparison. The earlier N4 B0 arm crosses below rho at $q = 4051$; the N4-v2 double-large-prime arm reaches 0.00523× negation-map rho at $q = 2{,}097{,}091$. This is a real restricted-family result. It is **not** a generic-curve, standard-curve, prime-order $E/\mathbb{F}_{q^3}$, or ECC2K-130 result.
3. **The best newly validated factor-base change is engineering progress, not a frontier crossing.** On the 20-bit prime holdout, the selected low-x base grows from $N = 32$ to $N = 45$ and costs 0.836, 0.817, and 0.746 of the incumbent at native-operation prices $w = 0, 0.1, 1$. It still costs 7.95, 8.24, and 9.62 times matched `rho.negation`. The result was reproduced on a second L2 EKS host, but covers one curve and gives no exponent. A later 19,887-job structural screen of Dickson, geometric-progression, and quadratic-progression bases found no useful growth in the coincidence multiplier: six five-rung slopes lie between -0.0047 and 0.0068, with the largest upper 95% limit 0.0147 versus the frozen 1/3 break-even slope. That narrows the factor-base search, but it prices neither E2E IC nor rho and therefore moves no ranked row.
4. **F5 and F6 improve local solver work without improving E2E ECDLP cost on the historical experiment binary.** F5 uses 0.641–0.647 of matrix F4's word work at degree 13. F6 uses roughly 0.77–0.82 of inherited-F4 word work and shortens the relation phase by 1.7–1.9×, but its charged closure/addition work makes its whole-pipeline lower-bound cost 64.6–1,352.6× inherited F4 and 399–4,883× rho on the tested cells. A fresh current-binary all-replay audit reproduces only 72/144 rows, so these numbers remain bound to the original measured binary.
5. **The optimized rho comparison is materially stronger at the implementation level.** The trajectory-identical signed-Frobenius rho path uses 0.802155 of the same-binary legacy Callgrind instructions, a 1.246642× speedup, while counted GAE and all outcomes remain identical. Its native wall improvement remains unpromoted because no L2 wall session completed. The new campaign therefore uses accounting-v2 rho, a same-target A/A rho control, and also measures BSGS and kangaroo where they can be finite-size leaders.

This report synthesizes versioned evidence; each performance row remains bound to the source and binary that produced it. The October snapshot executable audit is at source `ff7ad61cfed11908fb79de9dd485b696d0ada779`. It does not present blocked jobs as measurements. The exact current-source image is published, but the mandatory checkpoint gate has no upload credential. The replacement workload is now split into 92 one-hour-safe planning shards and all 92 pass the exact binary's planner. The deterministic analysis driver is hash-bound and passes its check-only gate, but exact-image runtime preflight, prospective freeze, and a current paid-execution identity are still missing. Consequently, there is no fresh 2026-10-31 wall-time leaderboard.

A follow-up executable establishment audit passes 19/19 artifact-consistency checks. It recomputes the decisive ratios, verifies all 30 retained Callgrind files, reproduces 720/720 optimized-rho rows with the current binary, and retains a new F4/F5/F6 compatibility failure (72/144 rows reproduce). A pass means the versioned evidence statements agree; it does not turn the draft EKS campaign into data.

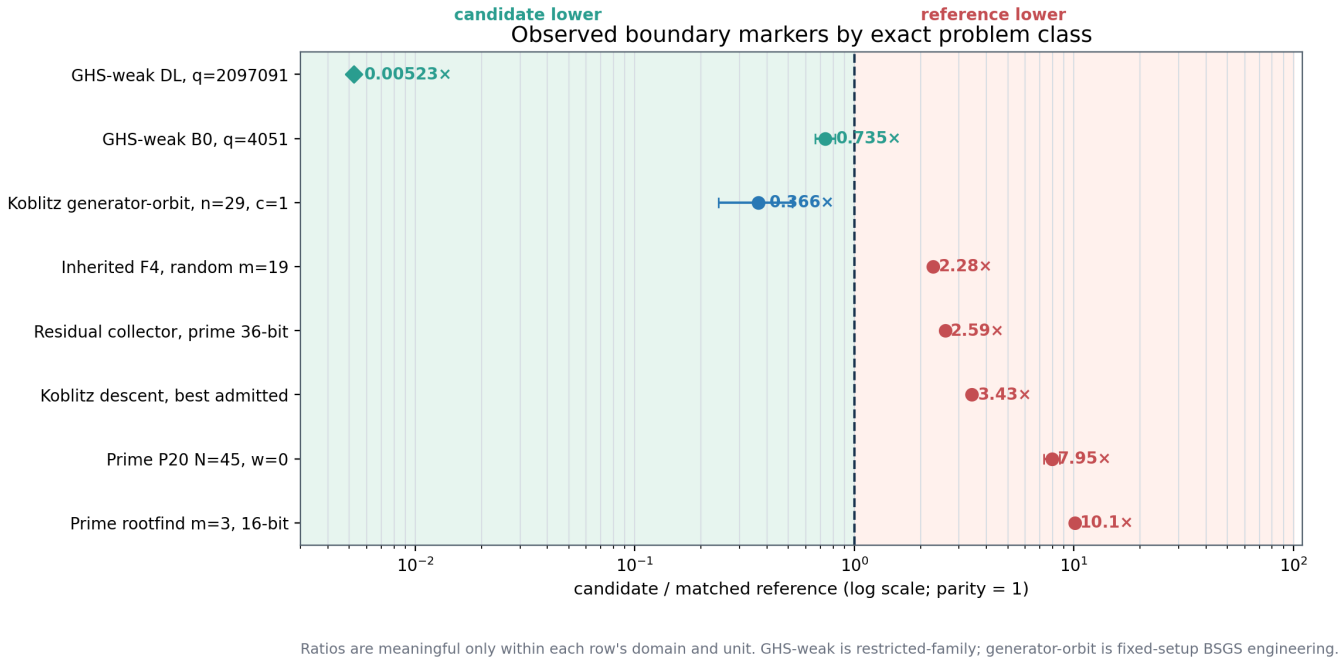


Figure 1: Boundary markers. A ratio is meaningful only inside its row's exact domain and unit.

2 1 November 78-method snapshot addendum

The live registry has grown since the pinned October snapshot. The current release binary at source `907d68631e79` has **78 top-level method IDs** plus **24 explicitly audited nested engines**: 11 system solvers, nine PDP3 engines, and four prime PKM tower engines. A fail-closed optimization audit admits 111 development specs with 78/78 methods and 24/24 nested algorithms; fresh local sessions exercised all 72 runnable native methods and every nested engine. The remaining six top-level entries are external CM wrappers without a real hash-bound backend.

The exact release binary SHA-256 is `e91c0ec79774af70c3b6af6aaac7b209e8edcdc0e42b1d18e1a92a14af065f65`. Across 35 admitted L0 sessions it audited 1,005 records, 952 verified outcomes, and reproduced all 782 eligible deterministic measured runs with zero problems. These are diagnostic development sessions, not L2 holdouts, so no leaderboard row changes.

The fresh, current-source performance boundary is:

- Koblitz n=29 generator-orbit IC is the only new holdout candidate, at 0.4405 [0.363, 0.534] of signed-Frobenius rho in its one-target, full-price L0 cell. This remains a fixed-setup candidate, not a family-level or leaderboard win.
- Pair-index F6 reduces inherited-F6 work to 0.2792 [0.246, 0.317], yet remains 139.30 [85.48, 227.02] times strong rho end to end. The prime PKM F4, F5, F6, and enumeration engines remain 4.62–5.03 times negation rho. The best matched n=17 system solver is wide FES/Gray-code at 4.28 times strong rho; SAT-CDCL, Crossbred, exhaustive, F4 variants, F5, F6, and all root-finding variants remain rho-led in their measured cells.
- The separate `ic bench` chained-x diagnostic is also rho-led: inherited F4 costs 24.71, 144.09, and 794.37 times rho at m=2, 3, and 4. Matrix F5 cuts matrix F4 work at m=3 to 0.6276, but still costs 16,296.76 times rho; both m=4 matrix arms censor because this oracle ignores the solver wall budget.

Mainline also now contains a distinct end-to-end GHS study, outside ECBench. It measured 30 applicable toy instances at N=12..28 and verified IC, rho, and BSGS on all 600 planted logs. Every measured instance remains generic-led: the best IC/best-generic ratio is 1.91 [1.72, 2.11], and the best IC/rho ratio is 1.60 [1.25, 2.12]. The genus-4 n=4 fit crosses near N=32 [31, 33], but that is extrapolated one step beyond the measured range. This does not invalidate the older `E_0/F_(q^3)` result in its exact restricted family and accounting domain; it does show that the historical exception cannot be generalized to the new GHS roster. The new GHS study has no ECBench replay certificate and is not leaderboard evidence.

Two later mainline packages extend the negative/control side of the frontier. The skeptic-reviewed evaluation of 12

prime-field IC leads reports no exponent movement. Sparse, dense, and Bochum-style factor bases match size-matched random nulls within about 0.0003–0.005 at the tested sizes. A characteristic-2 SAT gadget has conflict slope about 2.58 against the 0.5 m=3 ladder bar and is 3 times slower in CPU at its largest cell. Per-adder multiplier encoding is 0.78 [0.67, 0.92] of schoolbook conflicts at one b=9 prime, but reverses at b=6 and is censored at larger sizes. Clause retention moves a large setup into advice, and the proposed ring-linear propagator was not run. The surviving exponent questions are PKM at m>=4 and the contractor exponent; none of these rows is an ECBench speed claim.

The subsequent PFDR structured-base census adds 19,887 deterministic jobs on Dickson, geometric-progression, and quadratic-progression x-bases over 16–32 bits. All six family/theta lines pass the self-test and C3–C6 controls, and all read `O-SUB-BE`: their fitted log2(kappa)-on-log2(N) slopes range from -0.0047 to 0.0068, with no `ALIVE` line. This is strong negative structural evidence against those base families in the tested envelope. It is not an ECBench session, does not price a complete IC pipeline, and has no matched rho arm, so the ranked leaderboard remains unchanged.

R7.4 provides a separate binary-Edwards structural control. It proves that symmetrising by rational 2-power torsion gives the same algebraic system after an F2-linear substitution and checks 11 identity families on 46 toy curves with zero exceptions. The pair-table quotient sweep returns exactly the same relations with 1/mu lookups in all 268,000 checks. Its toy finite-size yield can lie on either side of parity depending on whether c or c+1 columns are charged and trends toward the null at n=13,14. No Groebner/F4 oracle or matched end-to-end rho run was performed, so this is not a new performance row.

2.1 Late-main 79th-method reconciliation

After the 78-method snapshot, main added `ic.prime_orbit_relations_v1`. The registry-wide optimizer now has an explicit finite-grid policy for its three true knobs (`coordinates`, `orbits`, and `relations_per_orbit`) and keeps both operation caps fixed as censoring/resource envelopes. The new fail-closed receipt covers **114 development specs, 79/79 top-level methods, and 24/24 nested algorithms**: 11 system solvers, nine PDP3 engines, and four prime PKM tower engines. It reports no missing entries, no structural errors, and no development/holdout seed reuse. The two warnings only restate that clocks in two older development specs are descriptive below L2.

The exact executable for the new sessions is source `68a78284` with release binary sha256:e6c1f5048acb071192f83ab9 96fd9577c152f136e8e62a301e50b62335ec7879. Three prospectively specified L0 sessions add 90 records, all verified, and all 72 measured deterministic executions replay identically. Cumulatively, the registry workstream now has 38 admitted L0 sessions, 1,095 records, 1,042 verified outcomes, and 854/854 eligible deterministic measured replays with zero audit problems. All 73 runnable native methods and all 24 nested engines have executed; the other six top-level methods remain external CM wrappers without a real hash-bound backend.

On the one-target generic development grid, affine coordinates with eight factor-base orbits and 1.5 requested relations/orbit use 0.8721 [0.754, 1.008] of the charged operations of the 16-orbit affine baseline. At 16 orbits, Jacobian uses 1.3351 [1.229, 1.450] times affine. This selects the eight-orbit arm for further testing, but does not justify changing the shipped default from one toy target.

The ECDLP comparison remains rho-led:

prime shape	tuned orbit IC / matched rho	pricing-band ratios at w=0,0.1,1	reading
generic a=-3	5.4169 [3.389, 8.658]	5.4169 / 5.0235 / 3.0548	negation rho leads
j=0, b=7	4.2011 [3.085, 5.721]	4.2011 / 4.0005 / 2.8097	cubic rho leads
j=1728	5.7901 [3.824, 8.768]	5.7901 / 5.4343 / 3.5160	quartic rho leads

These intervals are indicative within-target operation intervals, not curve-family generalization. Factor-base draws remain unpriced, making the IC totals lower bounds. Every run is L0 and each comparison has only four measured pairs, so local wall clocks are descriptive and no leaderboard row changes. The machine-readable evidence is the 79-method audit, generic comparison, j=0 comparison, and j=1728 comparison.

2.2 Boundary by domain

Domain	Current finite boundary	Leader	Evidence reading
Prime-field generic ECDLP	closest recent IC marker 2.59× matched rho; P20 validated base row 7.95–9.62×	generic algorithms	rho remains lower; BSGS can beat rho on several toy bitop cells
Random-binary generic ECDLP	inherited F4 2.28× rho on the tiny m=19 diagnostic; admitted ladder 10.6–38.2×	generic algorithms	no IC win
Koblitz generic ECDLP	admitted medium/scaling cold IC 3.43–11.5× matched rho; separate n=29 one-column fixed-setup ratio 0.366 [0.241, 0.523]	signed-Frobenius generic walk on the scaling object	no uniform-target/scaling IC win; fixed-setup BSGS engineering crossover retained separately
Special GHS-weak $E_0/\mathbb{F}_{q^3}$	B0 0.735× at q=4051; N4-v2 DL 0.00523× negation rho at q=2,097,091	plane-quartic IC	restricted-family IC win
Hyperelliptic Jacobians	genus-4 toy row 0.114× rho in bitop/v1	Jacobian IC	not an elliptic-curve result
Standard cryptographic curves / ECC2K-130	no complete break or scaling evidence	generic security boundary	unresolved by experiment; no claim of a break

3 What "the frontier" formally means

3.1 Domain key

A benchmark row is assigned to the exact domain

$$D = (\mathcal{C}, G, \mathcal{T}, A, k, U, P, E),$$

where $\mathcal{C}$ is the curve family and representation, G is the subgroup and cofactor treatment, $\mathcal{T}$ is the target distribution, A is reusable advice or preprocessing, k is the number of targets charged together, U is the accounting unit, P is the native-operation pricing rule, and E is the evidence class. Two observations from different domain keys do not form a speed ratio. In particular:

- prime GAE, binary word XORs, SAT conflicts, FES Gray-code vector XORs, and native wall seconds are not interchangeable;
- a one-target cold run and a retained-setup online run are different objects;
- a transfer to a special weak curve family is not evidence about the original generic family unless an explicit, subgroup-preserving, costed map is part of the run;
- L0 shared-host wall time cannot displace L2 isolated wall evidence.

3.2 Cost vector and dominance

For an admitted observation x in a fixed domain, retain the vector

$$C(x) = (W_0, W_{0.1}, W_1, T_{L2}, M, A, S, F),$$

where W_w is counted work with each unpriced native operation charged at $w \in \{0, 0.1, 1\}$ GAE, T_{L2} is isolated wall time, M is peak memory, A is reusable advice, S is success/censoring, and F is the retained failure set. Observation (x) dominates (y) only if both have the same domain key, (x) is no worse on every required axis, and it is strictly better on at least one axis. The frontier is the non-dominated set.

Unknown quantities remain unknown. They are never inserted as zero. A smaller backend-local counter can move a component frontier without moving the ECDLP frontier.

3.3 Admission gate

The refreshed ranking requires all of the following:

- exact curve, subgroup, target, factor-base, binary, and algorithm identity;
- cold one-target cost as the primary object;
- strongest applicable rho reference with the same usable symmetry, plus a same-target rho A/A control;
- verified scalar recovery and replay of every measured record;
- at least eight public targets and at least five measured rounds;
- at least four sizes before fitting or describing scaling;
- failures, timeouts, work limits, and censored pairs retained;
- the $w = 0, 0.1, 1$ pricing band whenever native operations are not fully converted;
- L2 or better isolation for a wall-time claim;
- no cross-ranking of backend-native units.

The full rules and machine-readable design are in PROTOCOL.md and protocol.json. The protocol validator reports zero findings, but the document remains `draft` until the exact current-source ECR identity, closed roster, and one-hour shard list are committed and frozen before execution. The image digest itself is already published and recorded.

4 Implementation census

The release binary exposes **74 top-level method IDs**:

Family	Count
Index calculus and algebraic/transfer candidates	31
Rho and transfer references	32
BSGS	7
Kangaroo	2
Pair-claw	2

The algebraic pipeline separately exposes **11 nested solver backends**: `f4-f2`, Buchberger, matrix F4, matrix F5, inherited F4, Crossbred, XL, SAT-CDCL, scalar FES, wide FES, and exhaustive search. These are not eleven additional top-level ECDLP methods; they are solver choices inside a complete pipeline.

Every ID and backend has an explicit disposition in COVERAGE.md and coverage.json: 13 current-primary methods, 10 diagnostic methods, 21 legacy controls, 6 pricing-sensitivity methods, and 24 domain-specialized methods. IMPLEMENTATIONS.md closes the additional factor-base, oracle, solver-engine, relation-matrix, target-source, and symmetry axes hidden behind those IDs. "Legacy" does not mean incorrect, and "blocked" does not mean slow; it records why the method is not in the fresh primary campaign.

The exact native registry is preserved in method-registry.json. This prevents "all implementations" from silently meaning only the arms that appeared in an older report.

5 Generic prime-field frontier

5.1 Generic algorithms are the finite-size leaders

On the 16–40-bit prime `bitop/v1` ladder, rho, BSGS, kangaroo, the pair-table IC rule, and the m=3 root-finding rule are measured on public synthetic prime-order curves. BSGS is below the current rho implementation on many toy cells (for example 0.558× at 16 bits), while rho approaches the better scaling boundary. The IC pair rule ranges from 6.23× to 78.3× rho; root-finding ranges from 10.1× to 69.7× through 28 bits.

This is why the draft campaign includes `bsgs.negation_v2` and `kangaroo.vow_v2` beside `rho.negation_v2`. Rho remains the asymptotic generic reference; it is not assumed to win every tiny finite instance.

5.2 Factor-base selection

The 2026-10-29 study jointly selected the low-x base size and exact pair oracle on training curves, then decided each size on a fresh holdout curve. The pair oracle stayed fixed; only (N) changed.

Bits	Incumbent → selected N	selected/incumbent at $w = 0/0.1/1$	selected/rho at $w = 0/0.1/1$	Decision
20	32 → 45	0.818 / 0.798 / 0.726 on frozen holdout; 0.836 / 0.817 / 0.746 on second host	7.95 / 8.24 / 9.62 on second host	promote N=45 engineering baseline
24	162 → 115	0.902 / 0.940 / 1.126	12.44 / 12.92 / 15.19	pricing relabelling; keep N=162
28	407 → 288	0.929 / 0.974 / 1.193	20.25 / 20.98 / 24.33	pricing relabelling; keep N=407
32	1024 → 1024	1 / 1 / 1	33.30 / 33.18 / 32.61	null selection

For P20, the second host admitted all 2,016 executions at L2, verified all 2,016, and matched 1,008/1,008 original records in every counted and band-priced field. The independent validation is filed in `VALIDATION.json`. The leaderboard's stale "no validation exists" sentence was corrected at the generator and generated Markdown/HTML/JSON surfaces.

The operational interpretation is narrow: increasing (N) helps the 20-bit pipeline because the improved relation probability pays back the larger base and table over that finite envelope. At 24 and 28 bits the decision changes with the native-operation price, so those candidates do not dominate the incumbent.

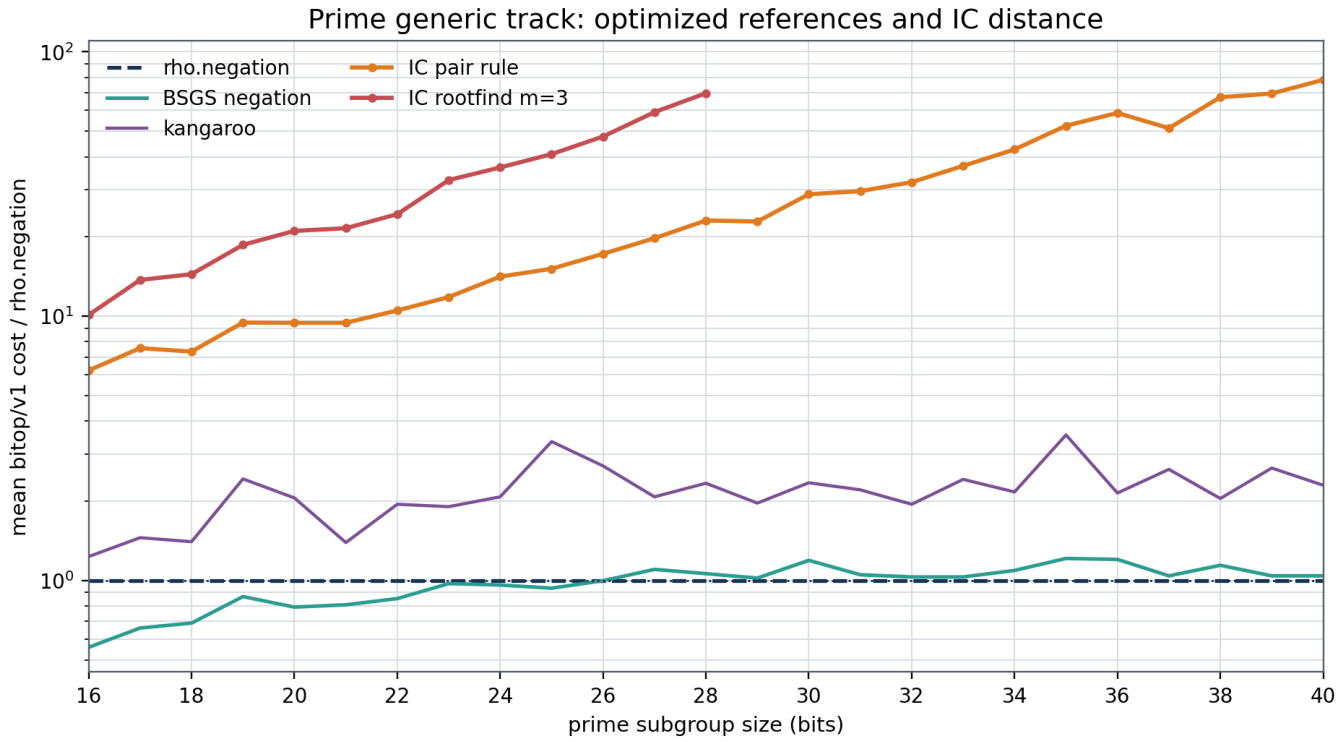


Figure 2: Prime finite-size ladder. BSGS is included so the rho reference is not mistaken for the absolute finite-size leader.

5.3 Structured factor-base growth screen

The preregistered PFDR census tests a different question from the low-x size sweep: whether structured x-base families make the coincidence multiplier κ grow with problem size fast enough to reach the frozen break-even slope $\delta^* = 1/3$. It ran 19,887 deterministic local jobs across 30 cells, with five rungs from 16 to 32 bits. The null calibration and planted-growth controls detect the slopes they are designed to detect; all blocking controls C3–C6 pass.

structured x-base	theta	fitted slope	95% interval	reading
Dickson	0.30	-0.0047	[-0.0130, 0.0041]	SUB-BE
Dickson	0.35	-0.0014	[-0.0088, 0.0070]	SUB-BE
geometric progression	0.30	0.0054	[-0.0031, 0.0136]	SUB-BE
geometric progression	0.35	0.0024	[-0.0049, 0.0098]	SUB-BE
quadratic progression	0.30	0.0066	[-0.0017, 0.0147]	SUB-BE
quadratic progression	0.35	0.0068	[-0.0004, 0.0144]	SUB-BE

This closes a tested-family branch, not the factor-base problem. It does not cover the absent FAM-1 MITM engine, a proper-subgroup arm, sizes above 32 bits, or any E2E cost. The original study is source-linked here.

5.4 Root finding: useful algorithm, wrong asymptotic label if overclaimed

`rootfind-subtract` and the m=3 root-finding arm are exact generic table/search procedures. They can be optimized and ranked E2E, but they do not receive the factor-base equations inside an algebraic system. They therefore belong to the **generic decomposition track**, bounded by the square-root frontier, rather than serving as evidence for a sub-square-root algebraic attack. The repository's 16–28-bit measurements remain 10.1–69.7× rho in `bitop/v1`.

The older root-finder replay audit also records an open regression: only 72/168 current-binary replays were identical on 2026-10-19. Affected historical stage counters remain blocked from promotion. The fresh campaign measures the

current implementation from new records rather than inheriting those counters.

5.5 Best recent prime marker and the missing algebraic baseline

The progress ledger's closest recent prime IC marker is the residual collector at 36 bits, 2.59× `rho.negation` in cold one-target (S). It is closer than the P20 factor-base study, but it is not a completed prime Gröbner-basis attack.

There is still no native prime-field ECBench arm that places the factor-base constraint inside a Semaev system and then runs a complete F4/F5/root-extraction pipeline against matched rho. Prime PKM/tower and GLV studies contain useful component counts, but their units and systems do not form that missing E2E baseline. This is the largest algorithmic coverage hole in the prime program.

5.6 Standalone GPU explicit relations

The RTX PRO 6000 study completes both fixed public 62-bit fixtures, with cold times 2,139.88 and 2,187.94 seconds in its V11 panel. Certificates replay and corruption controls pass. It remains **unranked** here because it is a standalone GPU implementation, its clocks are L0, and there is no same-fixture ECBench rho arm. It establishes an operational capability boundary, not a generic or matched-rho frontier.

6 Binary and Koblitz frontiers

6.1 Random binary

The admitted random-binary ladder remains 10.6–38.2× matched negation rho. In the newer F4/F5/F6 cross-family diagnostic, inherited F4 is closer on the tiny cells: 5.89× at m=13 and 2.28× at m=19. Two sizes do not establish an exponent, and the study's (S) is a lower bound because backend-native work is unpriced.

The key result is negative but useful: Koblitz-specific structure does not automatically improve the decomposer if the factor base is folded only by negation. The same solver ordering and similar local ratios appear on Koblitz and random-binary systems.

6.2 Koblitz

The admitted cold, one-target Koblitz table covers subgroup sizes around $(2^{\{36.6\})-(2}\{47.2\})$. IC/rho ranges from 3.43× to 11.5×. Setup accounts for roughly 98.7–99.6% of the cold IC pipeline on the central rows, so an online-only view can look attractive while the primary cold object remains rho-led.

The fixed-setup generator-orbit panel is a real but narrower exception. At n=29 and one factor-base column, `mitm-frobenius-layout-counted` with `stride-generator` has S=0.546/0.611/1.196 across the pricing band, versus 1.569 for rho-frob, and a paired ratio 0.366 [0.241, 0.523]. The same study shows why this is not the Koblitz ECDLP frontier: at one column the construction is Frobenius-folded BSGS, it uses a known generator-orbit column and a different target source, and its advantage comes from rho's fixed setup at log2(r)=15.4; the gap grows with r. It is retained as an engineering frontier rather than pooled with the medium/scaling uniform-target ladder.

The correct Koblitz reference uses signed Frobenius and negation. The canonical medium-tier counted boundary is 1.089× its floor, with interval [0.927, 1.261], across six sizes and 144 runs. Pair-claw is a useful memory/time frontier and reached 1.85× matched rho on one earlier cell, but did not displace the walk.

The factor-base symmetry lesson is asymmetric:

- folding a base or pair table can reduce stored representatives and setup;
- the matched rho reference must receive the same usable automorphism group;
- a symmetry that helps only a representation stage is not an ECDLP speedup;
- orbit naming, canonicalization, and retained advice must be charged to both arms consistently.

6.2.1 N83 four-byte root-index loop

The newest N83 implementation loop compares the plain four-byte ID31 root index with a packed two-bit fingerprint sidecar (FP2), using the same public point, binary, K1107 base, and independently replayed scalar. Both alternating orders pass `PASS_PAIRED_FULL_REPLAY`:

order	plain / FP2 index build	plain / FP2 whole process	plain wall	FP2 wall
plain then FP2	1.001×	1.017×	154.066 s	151.435 s
FP2 then plain	1.493×	1.294×	250.171 s	193.284 s

The code path is therefore correctness-backed and FP2 was faster in both full pairs, but this is not a selection result. The shared macOS host reported large load changes between orders, two pairs are too few for a stable effect, and FP2 adds a side array. At the projected K5500 split it still saves 2.25 GiB against the five-byte layout but adds 0.75 GiB against plain four-byte ID31. Peak RSS, full K5500 work, and a same-point Tier-1 IC/rho panel remain open.

7 The restricted GHS-weak exception

The repository's only clear IC-led elliptic-curve domain is the special GHS-weak family $E_0/\mathbb{F}_{q^3}$ with $\#E_0(\mathbb{F}_{q^3}) = 4\ell$. The comparison is a whole cold one-target pipeline in $\mathbb{F}_q$-multiplication equivalents, with no subgroup enumeration hidden outside the interval.

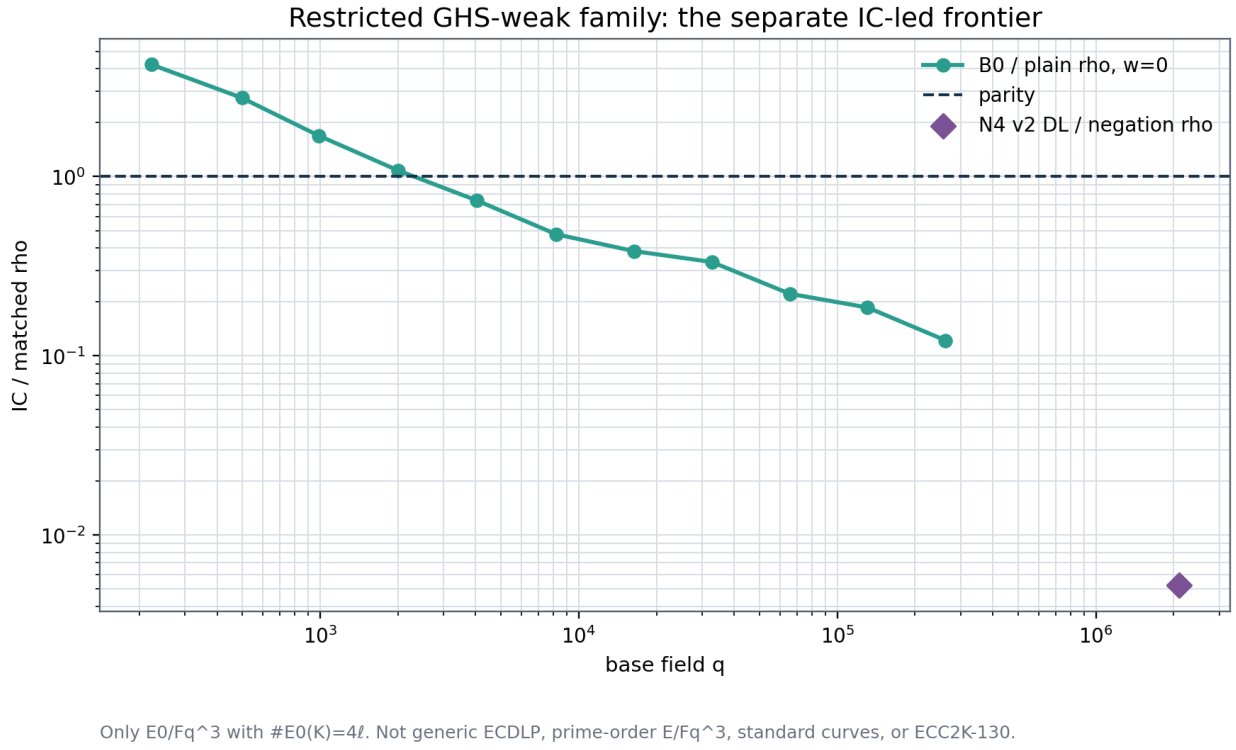


Figure 3: The GHS-weak curve family has its own, separate frontier.

The earlier N4 B0 arm crosses below plain rho at $q = 4051$, where its ratio is 0.735 [0.662, 0.816] at $w = 0$, and falls to 0.122 [0.0987, 0.157] by $q = 262111$. N4-v2 adds double large primes, structured Gaussian elimination, and block Lanczos; its DL arm reaches 0.00456× plain rho and 0.00523× negation-map rho at $q = 2{,}097{,}091$ in the recorded counted unit.

This result must travel with four restrictions:

1. the curves are intentionally GHS-weak and are not generic curves;
2. they are not generic prime-order $E/\mathbb{F}_{q^3}$ instances;
3. no standard cryptographic curve is in this family;
4. the result cannot be transported to another curve without an explicit, subgroup-preserving, costed map.

The correct synthesis is therefore not "IC never beats rho" and not "ECDLP is broken." It is: **IC beats rho on one special weak family; generic ECC remains rho-led in the measured envelope.**

8 F4, F5, F6, SAT, WDSAT, and FES

8.1 F4/F5/F6 result

The cross-family experiment uses eight targets in each of K13, R13, K19, and R19. Matrix F4/F5 at m=19 hit the retained 1,800-second timeout. At m=13:

Cell	Matrix F4 word XORs	Matrix F5 word XORs	F5/F4	Inherited F4 word XORs	F6 word XORs
K13	2.245e9	1.454e9	0.6465	1.112e7	9.10e6
R13	2.325e9	1.495e9	0.6414	8.66e6	7.99e6
K19	timeout	timeout	unknown	6.70e8	5.19e8
R19	timeout	timeout	unknown	7.15e8	5.62e8

These are historical-binary measurements. The 2026-10-31 current-binary `verify --replay-all` audit preserves every failure: K13 reproduces 16/48, R13 32/40, K19 16/32, and R19 8/24 deterministic rows. The X19 timeout control passes integrity but has zero replayable successful rows. In total, 72/144 rows reproduce and 72 differ; K13 differences include total GAE, phase GAE, and unpriced accounting. Stored record integrity and original verification still pass. Therefore the table establishes the ordering for the measured October 11 binary, not for the current implementation.

F5's signature criterion removes about a third of matrix F4's elimination work on the same tree. Inherited F4 is roughly 199–222× cheaper than matrix F4 on the m=13 systems. F6 further reduces word work and relation wall time, but performs exact factor-base point additions for its closure, which dominate the common priced pipeline.

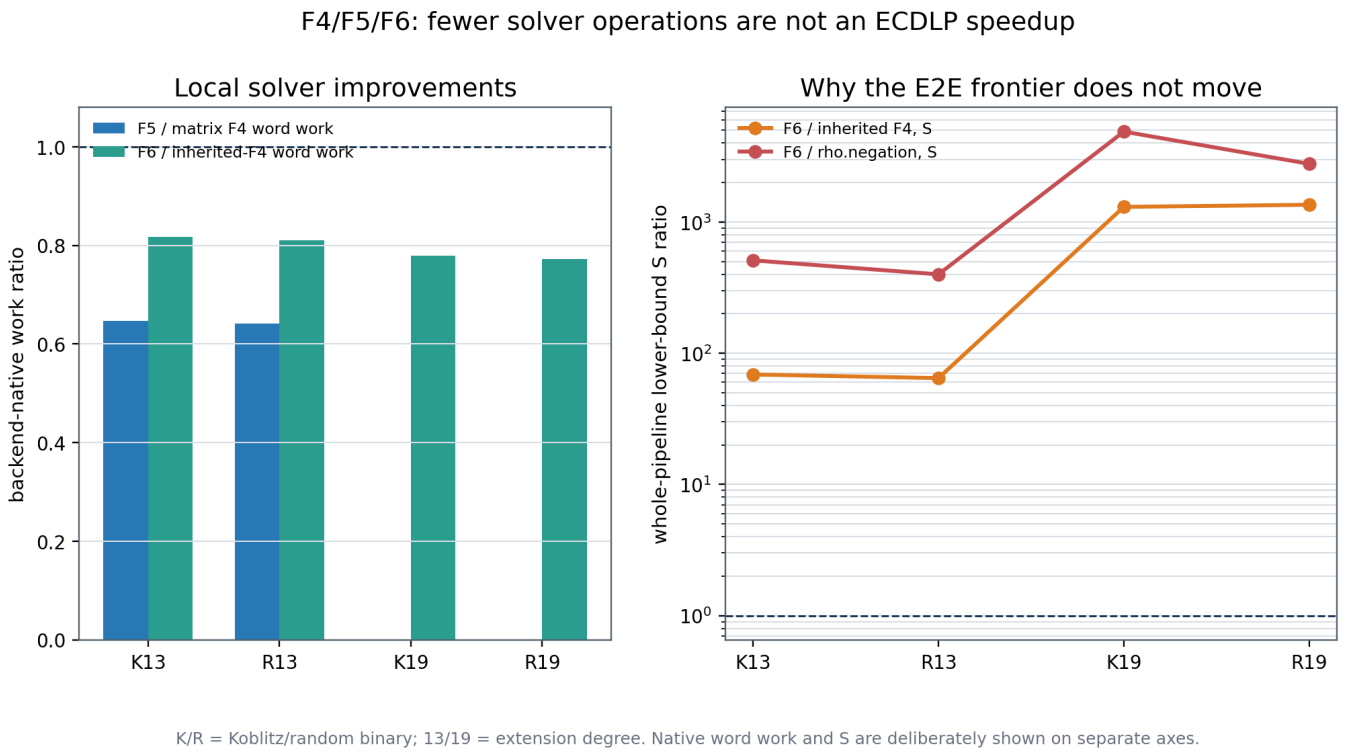


Figure 4: Backend-local savings and whole-pipeline cost move in opposite directions.

The versioned frontier rule follows directly: on that measured binary, F5 and F6 are component leaders in their native scopes, while inherited F4 is best of these four in the common E2E lower-bound (S), and none beats rho. A new current-source ranking requires the prepared all-backend rerun.

8.2 SAT-CDCL

SAT-CDCL is live in the nested registry and supports CNF-expanded parity and native-XOR parity. Existing n=17 records show why conflict counts cannot be a universal unit: native parity reports thousands of conflicts while CNF parity reports more than a million on related systems, with different clauses, auxiliaries, and encodings. A conflict reduction is a solver diagnostic unless CNF construction, XOR handling, relation yield, verification, and the whole pipeline share a common priced boundary.

The new solver-backend spec pins encoding and budget and runs SAT beside the same curve, factor base, target seeds, rho reference, and A/A control. Until that L2 session completes, SAT is implemented but not a ranked E2E leader.

8.3 WDSAT

WDSAT is a historical external path rather than a current native ECBench method ID. XG/Gaussian-elimination mode can cut its own conflict count by tens or hundreds of times, but "conflict" changes meaning with solver, encoding, and XG mode. More importantly, a retained audit found a false-UNSAT result on one of sixteen decomposable targets in the unfixed path. The historical WDSAT evidence is therefore useful for regression and solver research, not for promotion to the current E2E frontier.

8.4 FES and Gray code

`fes-f2` uses scalar Gray-code enumeration; `fes-f2-wide` evaluates multiple subcubes through AVX2/AVX-512 and is the strongest exhaustive reference where its equation limit applies. Gray-code word XORs and vector XORs are native units, not GAE. Past component studies show real implementation gains, but a universal gate was not cleared on every cell. The fresh all-backend spec keeps both implementations, their timeouts, and their native counters, while ranking only common E2E outcomes.

8.5 Crossbred, XL, Buchberger, and exhaustive controls

These backends remain important controls:

- Buchberger checks that matrix techniques are not winning by an incorrect algebraic shortcut;
- Crossbred tests hybrid specialization/elimination;
- XL provides a degree-lifted linearization baseline;
- exhaustive search supplies the correctness and small-system floor.

Their native counter definitions differ, so no report should divide one by another and call the quotient a speedup. The common comparison starts at the same ECDLP target and ends only after the recovered scalar verifies.

8.6 Tensor-contraction rank boundary

The hashed-ECDLP tensor branch now has a rigorous architecture-level boundary, not a performance promotion. `tensor-census` schema v2 emits an exact nonzero minor for the canonical and selected balanced flattenings; the verifier reconstructs the boundary matrix and recomputes the determinant independently with fraction-free Bareiss elimination. A 1,152-matrix deterministic control agrees with rational Gaussian elimination and rejects a changed determinant.

Lean proves that a nonzero $k \times k$ minor gives rank at least k, and that an exact linear separator factoring the matrix through d coordinates has rank at most d. Thus the finite certificate forces $k \leq d$ for that exact linear-message architecture. The asymptotic work theorem is explicitly conditional: no uniform exponential-rank lemma is yet proved for the production hashed prime-field family, and no production E1 certificate exists. The result rules out no nonlinear, randomized, approximate, alternate-encoding, or separator-avoiding algorithm and changes no ECBench row.

9 Rho comparison: current optimized boundary

The optimized `rho.signed_frobenius_strong` path preserves the exact walk, recovered scalar, counters, and GAE while reducing implementation instructions:

Measurement	Result
Optimized / same-binary legacy Callgrind instructions	0.802155
Optimized / archived pre-change instructions	0.804582
Speedup over same-binary legacy	1.246642×
Canonicalization share	53.91% → 42.48%
Verified executions	864 / 864
Measured replays identical	720 / 720
GAE optimized / legacy	1.0000 [1.0000, 1.0000]
Native L2 wall verdict	unknown

The implementation win comes from paired coordinate maps, fixed common byte shapes, and shared coefficient reduction. It does not change the mathematical rho operations boundary. A fresh audit hash-checks all 30 raw Callgrind files and independently recomputes 1,995,304,985 legacy versus 1,600,543,701 optimized instructions; the current binary also reproduces all 720 measured deterministic rows. The old L0 wall ratio is not promoted.

For the new campaign:

- prime uses `rho.negation_v2` and a byte-identical A/A arm;
- Koblitz carries both `rho.signed_frobenius_strong_v2` and `rho.signed_frobenius_strong_escape`, with the former duplicated for A/A and the reference-selection rule frozen before data;
- BSGS and kangaroo are measured on prime cells to reveal finite-size leaders;
- every symmetry available to an IC arm is audited for the reference;
- setup, distinguished-point policy, table state, restarts, and verification stay inside the declared interval.

10 Bottleneck map

Bottleneck	Evidence now	Why it holds the frontier	Required next gate
Relation probability / trials	prime and binary IC gaps grow across several ladders	solver improvements do not change the probability of a useful decomposition	algebraic factor-base constraint inside the system; four-size E2E sweep
Structured factor-base yield	six PFDR lines over three families and 16–32 bits have slope -0.0047 to 0.0068; largest upper 95% limit 0.0147 versus 1/3	the tested structures do not make useful coincidences grow fast enough	test a new mechanism or integrate the base constraint into a fully priced E2E solver; do not extend these families by default
Factor-base and pair-table setup	98.7–99.6% of central cold Koblitz rows; $O(\|F\|^2)$ pair materialization	online-only improvements disappear when one target pays setup	cold primary result; equal-advice online secondary result
N83 root-index layout	two independently replayed K1107 pairs favor FP2 by 1.017× and 1.294× whole-process, but shared-host load changes sharply	the fingerprint trades reconstruction work for side-array memory; K1107 does not predict K5500 peak RSS or cost	larger-memory K5500 paired Tier-1 run with matched rho, fixed order policy, and peak-RSS receipt
Tensor contraction separator	exact finite nonzero-minor certificates and the linear-separator implication are proved; production family lemma absent	finite architecture bounds do not establish the rank growth or cover other algorithms	generate a production semantic cut and prove a uniform family lower bound before making an asymptotic claim
F6 closure additions	historical binary: 64.6–1,352.6× inherited F4 in lower-bound S; current replay 72/144 overall	local word-XOR savings are overwhelmed by exact group work, while current counts need remeasurement	price closure and solver in one common unit; rerun current source; change closure algorithm, not label
Matrix F4/F5 scale	m=19 matrix arms time out at 1,800 s	symbolic preprocessing and matrix construction explode before useful scaling	memory/degree receipts and retained timeout curve
Solver-counter mismatch	XORs, conflicts, vector steps, multiply-adds	cross-solver ratios lack semantic equality	E2E wall or complete common pricing; native counters remain diagnostics
Linear algebra	large stage-only speedups, small E2E changes in several pipelines	relation collection and setup dominate	measure phase share before further LA work
Rho reference quality	19.8% instruction cut; toy BSGS sometimes below rho	a weak reference creates false IC wins	current accounting-v2 rho, A/A, BSGS/kangaroo finite-size controls
Prime algebraic coverage	no complete native prime F4/F5 factor-base-constrained arm	generic root finding cannot answer the algebraic claim	implement and validate a prime algebraic reference arm
Evidence portability	mixed L0/L1 historical wall, one-curve studies	host effects and curve effects are confounded	L2 second host, at least 8 targets, 5 rounds, and 4 sizes

Bottleneck	Evidence now	Why it holds the frontier	Required next gate
GPU integration	verified 62-bit standalone result, no matched rho	capability result cannot enter the ECBench Pareto set	native adapter, same fixtures, matched reference, isolated clocks

The dominant pattern is that improvements are arriving after the point that currently controls total cost. Fast root extraction does not fix relation probability; fast linear algebra does not fix pair-table setup; fewer F6 XORs do not fix closure additions. The next useful optimization should be selected from the measured whole-pipeline phase share, not from the largest isolated microbenchmark speedup.

11 Curve-by-curve interpretation

11.1 Toy curves

Toy curves are valuable for exhaustive controls, replay, seed-law tests, solver parity, and detecting fixed overhead. They do not establish a security claim. At 16–26 prime bits, BSGS can beat rho in `bitop/v1`; at binary m=13, fixed setup can make the "strong" parallel rho variant worse than simple negation rho. The report therefore retains multiple generic controls.

11.2 Medium experimental curves

The repository's most informative generic boundary comes from ladders with enough sizes to reveal whether a fixed overhead is fading or a gap is growing. Prime pair/rootfinding costs move away from rho through 40/28 bits. Koblitz one-target IC remains several times rho through roughly 47 subgroup bits. The 62-bit GPU fixture is operationally larger but is not a matched frontier row.

11.3 Special transfers and covers

Isogeny, quartic-cover, GHS-cover, three-leads genus-2, and hyperelliptic paths must each retain the map, subgroup image, target transport, and inverse recovery cost. A cheaper DLP on the destination object helps only when those obligations are explicit and cheaper end to end. The GHS-weak family is a direct restricted problem class, not a free transfer from arbitrary curves.

11.4 Standard curves and ECC2K-130

No standard curve in the registry has a completed IC solve that changes its security boundary. The repository's ECC2K-130 material is a feasibility and solver program, not a demonstrated break. Claims about standard curves remain bounded by generic algorithms and external cryptanalytic knowledge, not by extrapolating toy ECBench fits.

12 Fresh campaign status

The original nine exact-build specs remain preserved as the broad planning draft: 4,704 processes, several per-child timeouts equal to the entire one-hour queue deadline, and missing domains. They will not be submitted.

The replacement contains 92 candidate/reference/A/A specs. Two four-target blocks give eight prospectively selected targets per comparison cell; each has one warmup and six measured rounds in balanced alternating order. A shard with a arms uses child timeout $\lfloor 2400/(4 \cdot 7 \cdot a) \rfloor$ seconds. The resulting range is 12–28 seconds, and the largest sum of timeout ceilings is 2,380 seconds. At least 1,220 seconds of the EKS deadline therefore remains for deterministic workload construction, Karpenter provisioning, setup, audit, and upload.

Panel	Jobs	Planned processes	Target law
Prime 20/24/28/32, IC and generic algorithms	16	2,632	public
Koblitz 13/19/23/31, rho/rootfinding/F4/F6/claw	32	5,152	public

Panel	Jobs	Planned processes	Target law
Random binary 13/19/23/31, matrix F4/F5 and inherited F4/F6	16	1,792	public
Eleven nested algebraic solver backends	12	1,960	public
j=0 and j=1728 symmetry	4	672	public
n=29 generator-orbit fixed-setup engineering	2	392	public
F_((2^m)4), specialized m=10/12/15 and wide m=16	8	1,008	planted known-answer
Binary n=11 double-large-prime v2	2	280	public

The exact pinned executable accepts all 92 specs, expands 13,888 processes, and reports zero planning failures. The closed roster check covers all 13 top-level methods classified `current_primary` and all 11 nested solver backends. Exact spec IDs and file hashes are retained in execution/shard-preflight-plans.json, and SHARDS.md gives the complete reference rules.

The extension-field target law is a disclosed limitation. Each `f2m4_torsion` fixture exhausted the harness's fixed 10,000-attempt public hash-to-subgroup constructor. Those eight specs therefore use planted known-answer targets, with the scalar hidden from every solver. They can compare implementations on matched targets but cannot enter the public-target frontier. The wide m=16 arm is also expected to censor under the short queue cap; historical executions took hours.

Trusted publication run 41241104247 built the exact static x86-64 binary, SHA-256 924d090c1d83a20ce954848c04b1 16cee4d5a7cb04187bb618dff60958bfd64d, and image digest sha256:b8c20f9c151360b78f505360e8d47068c4068f00a8f6 faa7ebd3086fa0a791c6. That run was publication-only; it submitted no experiment.

Paid execution remains blocked. `scripts/checkpoint.sh gate` is mandatory, but this workspace has no checkpoint upload token, token file, AWS CLI, or AWS identity. Repository authentication now succeeds for the authorized account, but no direct Iso Lab tool is installed and GitHub access alone cannot satisfy the checkpoint or execution-authority gates. The analysis driver is already hash-bound; exact-image runtime preflight and the prospective protocol freeze remain open. The exact state is in execution/BLOCKER.md. Local wall timing is not silently substituted.

Before the first warmup, the executor must:

1. provision an authorized AWS/Iso Lab submission identity while retaining the working repository identity;
2. run exact-image preflight for each arm class and preserve the bound analysis script;
3. provision checkpoint transport and pass `scripts/checkpoint.sh gate`;
4. freeze and merge these exact 92 spec IDs to `main`;
5. reconcile existing receipts and submit each paid shard exactly once;
6. upload raw records, host capsule, replay receipt, A/A analysis, and hashes;
7. retain every timeout, failure, and unattempted cell;
8. regenerate the Pareto tables and this report from admitted records only.

13 Conclusions and research priorities

The performance frontier is now explicit:

- **Generic ECC:** rho/BSGS/kangaroo remain the medium/scaling leaders; the n=29 one-column generator-orbit fixed-setup BSGS crossover is separate and does not establish a generic IC crossing.
- **Restricted GHS-weak ECC:** plane-quartic IC is the leader and must remain labelled as a special-family result.
- **Prime engineering:** P20 $N = 45$ is the validated factor-base baseline, still roughly 8–10× rho. The newer orbit collector's best toy configuration is affine/eight-orbit, but remains 4.20–5.79× matched rho across its three

L0 prime-shape checks. The PFDR screen also rejects useful coincidence-growth evidence for Dickson, geometric, and quadratic x-bases through 32 bits; it is structural evidence, not an E2E comparison.

- **Algebraic solvers:** F5 and F6 win component metrics; inherited F4 wins the tested common E2E lower-bound comparison; rho wins the ECDLP comparison.
- **SAT/WDSAT/FES:** implemented or historically explored, but not yet on one common, validated E2E frontier; WDSAT also carries a retained correctness failure in an old path.
- **Koblitz implementation loop:** the four-byte FP2 N83 root index passes two alternating full K1107 replays and is faster in both observed pairs, but remains unselected until a stable K5500 Tier-1 comparison prices its added side-array memory and includes matched rho.
- **Tensor contraction:** finite linear-separator rank certificates and their proof bridge are established; the production-family rank-growth lemma is still open, so no ECDLP or leaderboard claim follows.
- **Root finding:** valuable generic decomposition machinery, not evidence for an algebraic sub-square-root attack.
- **Rho:** instruction-level optimization is real and verified; isolated native wall promotion is still open.

The highest-value next work is therefore:

1. provision checkpoint/submission transport, finish runtime preflight, merge the closed shard protocol with its bound analysis driver, and execute its L2 campaign;
2. implement the missing prime factor-base-constrained algebraic arm;
3. attack relation yield and cold setup before further optimizing stages that are already a small share;
4. integrate the GPU relation engine with a matched ECBench reference;
5. run the paired N83 four-byte/FP2 K5500 Tier-1 memory and E2E gate;
6. prove or falsify the production tensor-family rank-growth obligation;
7. preserve the GHS-weak win as a separate domain rather than using it to headline generic curves.

14 Reproduction and evidence index

The report is generated only from committed data and prior frozen evidence. Presentation data are in frontier_snapshot.json, and `render_figures.py` renders all four figures. Important sources are:

- canonical IC leaderboard;
- ECBench contract;
- backend-frontier contract;
- F4/F5/F6 result;
- rho optimization;
- prime factor-base second-host replay;
- PFDR structured-factor-base screen;
- N83 four-byte paired root-index replay;
- rigorous tensor boundary-rank report;
- GPU explicit-relations result;
- literature and prior-evidence ledger;
- live coverage, closed roster, and queue-safe shards;
- campaign protocol and pre-data analysis driver;
- executable establishment certificate.

Primary mathematical references used to set the boundary include Shoup's generic-group lower bound (EUROCRYPT 1997, DOI 10.1007/3-540-69053-0_18), Faugère's F4 (DOI 10.1016/S0022-4049(99)00005-5) and F5 (DOI 10.1145/780506.780516), Semaev's summation-polynomial work (IACR ePrint 2004/031), and the Faugère–Perret–Petit–Renault binary-field IC analysis (DOI 10.1007/978-3-642-29011-4_4).

No sentence in this report converts a missing measurement into a loss, a local counter into an E2E speedup, a special-family result into a generic break, or a draft EKS plan into fresh benchmark data.